

Cost-Effective 448-Gbps Indoor Optical Wireless Access Via Dual-Polarization Carrier-Extracted Direct Detection

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Abstract *We propose a dual-polarization carrier-extracted direct detection without polarization tracking because of the polarization stability in free-space environment. The system achieves a 448-Gbps transmission data rate in 9-meter indoor free-space link without expensive local oscillators while meeting the eye-safety standard. ©2026 The Author(s)*

Introduction

The exponential growth in data traffic, driven by emerging applications such as augmented reality, 8K video streaming, and the Internet of Things, is pushing the capacity limits of conventional radio-frequency-based wireless indoor access networks [1]. While these systems continue to evolve, the scarcity of the radio spectrum and escalating interference issues present fundamental bottlenecks for achieving Terabit-level data rates in future indoor environments. Optical wireless communication (OWC), including visible light communication (VLC) [2] and near-infrared (NIR) communication, has emerged as a promising complementary technology to overcome this “spectrum crunch” [3]. By leveraging the vast, unregulated optical spectrum, OWC offers unparalleled advantages, including ultra-high bandwidth, inherent security, and immunity to electromagnetic interference. For indoor access scenarios, OWC can establish high-speed “optical wireless links” to deliver data directly to users, potentially revolutionizing the connectivity experience in offices, homes, and data centres.

Compared with VLC, NIR OWC is more compatible with optical fibre communication systems and easier to achieve high data rates [4]. Intensity modulation direct detection (IM/DD) is constrained by its low spectral efficiency, which limits its ability to achieve high-capacity indoor OWC. Coherent detection is indeed performant and easy to achieve higher speed in OWC [5] but it is quite costly. Therefore, self-coherent detection become a good candidate because it can realize full-field recovery without local oscillators (LOs) and relax laser requirements [6], [7], promising for cost-effective ultra-high-speed indoor OWC. Researchers have also demonstrated high-speed indoor OWC based on self-coherent detection [8], [9], achieving 200-Gb/s data rate [9]. To further enhance the capacity, polarization division multiplexing (PDM) is an effective

technology. While PDM self-coherent system always needs active polarization tracking to recover the transmitted PDM signal in fibre-optic system [10], the atmospheric propagation has few effects on the light polarization state even if the channel is turbulent [11], [12]. Therefore, the polarization state of transmitted PDM signal can be preserved to obviate the need for active tracking which is costly and complex [13].

In this paper, we propose and experimentally demonstrate a novel scheme termed dual-polarization carrier-extracted direct detection (DP-CEDD) based on a narrowband silicon photonic (SiP) integrated coupled resonator optical waveguide (CROW) filter for indoor OWC, which is one kind of self-coherent detection schemes. Benefited from the polarization-stable and weak-turbulence indoor environment, the receiver-side adaptive polarization tracking is not necessary for DP-CEDD. We successfully validate this scheme through a 448-Gbps transmission over a 9-meter indoor free-space link. This work represents the advantages of high capacity, low cost and high spectral efficiency for indoor OWC systems, underscoring its feasibility as a core technology for cost-effective, ultra-high-speed indoor optical wireless access.

Principles

As shown in Fig. 1(a), to ensure the successful PDM DD signal recovery, an adaptive polarization controller (APC) cannot be eliminated in the conventional IM/DD systems over the optical fibre transmission [14]. The APC is used to track and compensate the polarization wandering of the DD signal. While the signal and carrier are co-transmitted in the self-coherent systems, an APC is also necessary in the receiver side [10] as depicted in Fig. 1(b), increasing the power consumption and cost induced by a complex electrical control loop.

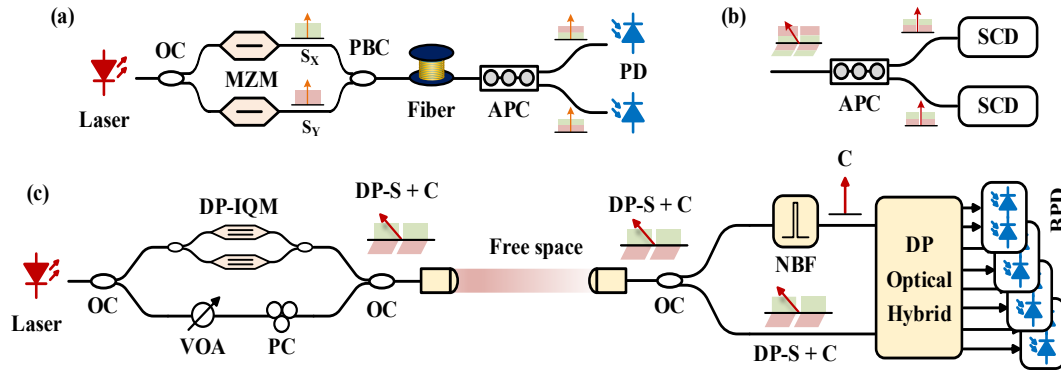


Fig. 1: (a) Conventional DP direct detection-based fiber-optic communication architecture; (b) Conventional DP self-coherent receiver scheme; (c) DP CEDD based free-space optical communication system scheme. OC: optical coupler. MZM: Mach-Zender modulator. APC: adaptive polarization controller. PD: photodetector. SCD: self-coherent detection. DP-IQM: dual-polarization IQ modulator. VOA: variable optical attenuator. PC: polarization controller. DP-S: dual-polarization signal. NBF: narrowband filter.

Given the polarization-stable free-space optical (FSO) channel, which preserves the polarization state of transmitted signals, PDM DD systems are inherently well-suited for such applications, as they eliminate the need for active polarization tracking loops. The proposed DP-CEDD scheme is represented in Fig. 1. (c). The modulated DP signal and optical carrier are co-transmitted. At the receiver end, the optical carrier can be extracted from the received signal by a narrowband filter to beat with the remaining signal in another path. As the pure carrier is extracted, the CEDD is similar to coherent detection in principles and the digital signal processing (DSP) is also simplified because the signal and carrier are from the same optical source. In contrast to conventional self-coherent detection techniques such as CADD [6] and Stokes Vector direct detection (SVDD) [7], the CEDD avoids the complex signal-to-signal beat interference (SSBI) cancellation algorithm and supports extension to PDM configurations. Notably, the active polarization tracking for DP-CEDD receiver becomes unnecessary in these indoor OWC scenarios, as plotted in Fig. 1(c). Only static polarization adjustment at the transmitter is required to ensure successful recovery of the DP signal. For practical

applications, the transmitter and receiver of DP-CEDD should be integrated into a single chip instead of a system based on fibre devices, further ensuring polarization stability. Therefore, DP-CEDD can be potentially a cost-efficient and high-capacity scheme, promising for high-performance and ultra-high-speed indoor OWC.

Experimental Setup and Results

Fig. 2 presents the experimental setup of the FSO DP-CEDD system. Light from a distributed feedback (DFB) laser is split into two branches via an optical coupler: one branch is fed into a dual-polarization IQ modulator driven by a Keysight M8194A arbitrary waveform generator (AWG) at 120 GSa/s, while the other serves as the optical carrier. The system employs 16 Quadrature Amplitude Modulation (16QAM)-modulated Orthogonal Frequency Division Multiplexing (OFDM) signals with a DFT size of 2048. Using 1434 subcarriers for data transmission, a raw data rate of about 448 Gb/s is achieved. A 4-GHz guard band is inserted within the modulated spectrum to facilitate pure carrier extraction. The modulated signal and optical carrier are combined before transmission. Prior to traversing the 9-meter indoor FSO link, the self-coherent DP

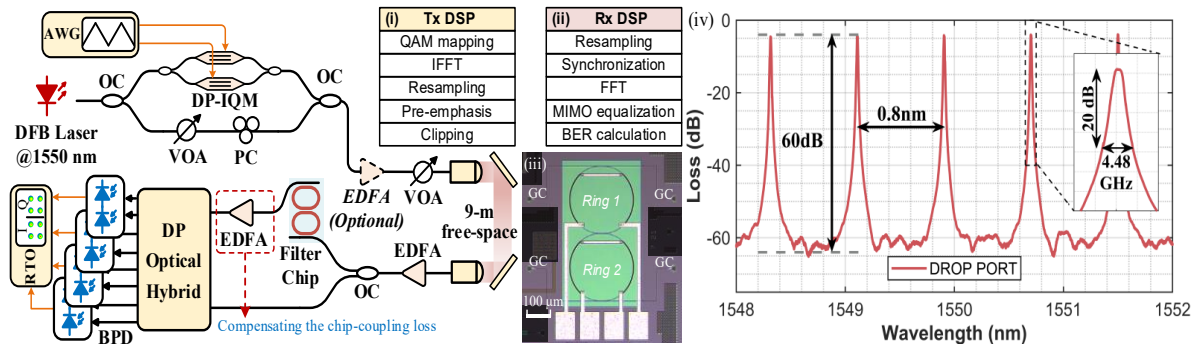


Fig. 2: Experimental setup of measuring FSO DP-CEDD system. Insets: (i) Transmitter-end DSP; (ii) Receiver-end DSP; (iii) Microscope image of narrowband filter for carrier extraction; (iv) Spectral response of the narrowband CROW filter with a resolution of 0.1 pm (the loss has been normalized by removing on- and off-chip coupling losses). EDFA: erbium-doped fiber amplifier. GC: grating coupler.

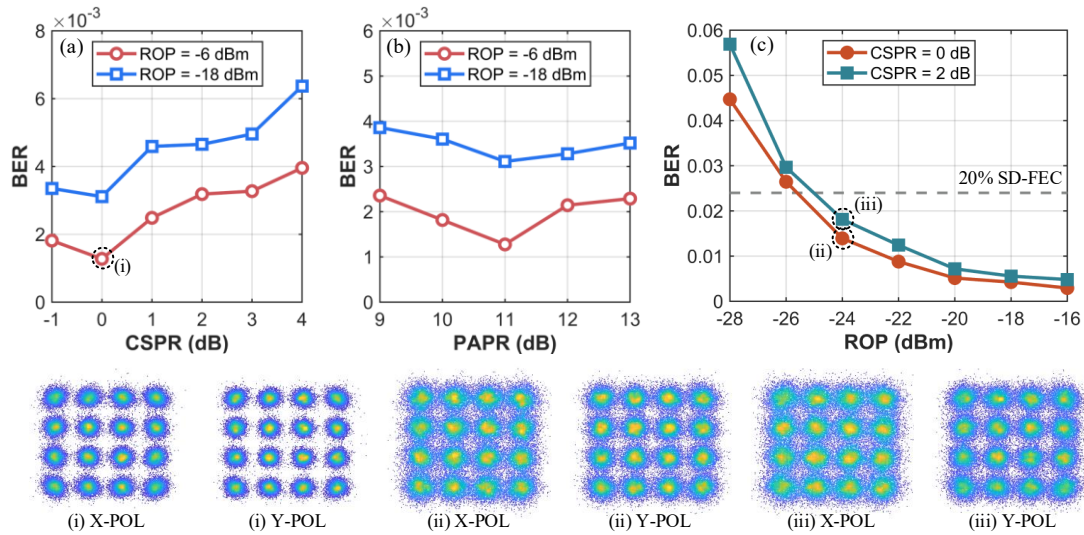


Fig. 3: BER performances versus (a) CSPR, (b) PAPR and (c) ROP. Insets: (i)-(iii) Recovered constellations of transmitted signals. CSPR: carrier-to-signal power ratio. PAPR: peak-to-average power ratio. ROP: received optical power.

signal, comprising both signal and carrier, is boost-amplified at the transmitter. The VOA is deployed here to attenuate the emitting power to 0 dBm to meet the eye-safety standard [4]. At the receiver, the signal is first pre-amplified by an EDFA and then split into two paths. One path is sent directly to a standard DP coherent receiver; the other passes through a narrowband filter for carrier extraction, followed by EDFA amplification to compensate for the on- and off-chip coupling losses (~ 11 dB). The EDFA for coupling-loss compensation can be eliminated when implemented with a fully-integrated receiver [15], [16]. The extracted carrier is also fed into the coherent receiver. Finally, the RF outputs are captured by a Keysight DSAV334A real-time oscilloscope (RTO) at 80 GSa/s for offline DSP. Figs. 2(i)-(ii) depict the CEDD DSP flow, and Fig. 2(iii) shows the microscope image of the SiP CROW filter used as the narrowband filter, which exhibits a 4.48-GHz 20-dB bandwidth and 60-dB extinction ratio as illustrated in Fig. 2(iv).

Fig. 3(a) shows the carrier-to-signal power ratio (CSPR) optimization results at -6 dBm and -18 dBm received optical power (ROP) respectively. The VOA deployed in the transmitter is applied to change the CSPR value. As there is a trade-off between effective signal power and desired carrier-to-signal beat terms, the CEDD system achieves best bit error rate (BER) performance when the CSPR is 0 dB in this experiment. In the CEDD system, the optimal CSPR value selection is also affected by the bandwidth and extinction ratio of the narrowband filter, which determining the purity of the extracted carrier. Fig. 3(b) illustrates the peak-to-average power ratio (PAPR) optimization results. In this experiment, we employ clipping to change the PAPR of the OFDM signal. Therefore, high PAPR enhances the

quantization noise in the digital-to-analog converter (DAC) while excessive clipping causes the signal distortion. When the PAPR value equals to 11 dB, the system can achieve the best BER performance.

To evaluate the eye-safety power margin of the DP-CEDD FSO communication setup, we further reduce the emitted optical power at the transmitter using the VOA. As shown in Fig. 3(c), under the optimal CSPR (0 dB) and PAPR (11 dB) and considering a 20% soft-decision forward-error-correction (SD-FEC) threshold [17], the CEDD system can achieve a ROP sensitivity of approximately -25 dBm. An ROP of -25 dBm corresponds to a free-space optical power of approximately -19 dBm, which is far lower than the eye-safety standard requirement of 10 dBm [4], [5]. With high sensitivity comparable to coherent detection, DP-CEDD offers a large power margin, making it a promising solution for indoor OWC applications.

Conclusions

In this paper, we propose a DP-CEDD system which employs a narrowband filter to extract optical carrier from the transmitted signal where a LO-free coherent-like detection scheme is realized. Taking advantage of the fact that the indoor FSO channel is short and stable which has few influences on the polarization state of transmitted signal, DP-CEDD system can eliminate its need for APC, reducing power consumption and cost. We demonstrate a 448-Gbps transmission data rate over 9-m indoor FSO channel based on this polarization-tracking-free DP-CEDD system with the advantages of high spectral efficiency, high sensitivity and low cost, which present its promising application potential in high-capacity and cost-effective indoor optical wireless access.

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